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HTML :

Module 1- Introduction

- Basics of html and document structure
- Doctypes
- Basic Tags

1. html
2. head
3. body
4. title
5. meta
6. link
7. script

Module 2 - Page Formatting

- Paragraphs
- Line breaks
- Blank space
- Pre Formatting text
- Headings
- Comments

Module 3-HTML Elements

- Element Types
- Block Elements
- Inline Elements
- Div
- Span
- Header
- Footer
- Section
- Article
- Nav
- Aside

Module 4 - Lists

- Un Ordered Lists
- ordered Lists
- Nested Lists
- Definition Lists

Module 5- Links in HTML

- What are links
- Text links
- Image links
- Opening link in new window / new tab
- Opening link in same same tab
- Links to content of same page
- Linking to an email
- Linking to a file

Module 6- Images

1. Introduction to images
2. Adding an image to web page
3. Image alt text
4. Image labels

Module 7-Tables

1. Displaying data in table
2. Table Header
3. Table Body
4. Table
5. Thead
6. Tbody
7. Tr
8. Td
9. Th
10. Row span
11. Col span

Module 8 -Forms

1. Introduction
2. Text Box
3. text
4. password
5. email
6. number
7. date
8. range
9. url
10. color
11. Radio Button
12. Check Boxes
13. Select Box
14. Multiple Select Box
15. Text Area
16. Submit Button
17. Reset Button

Module 9 -Storage

1. Web Storage
2. Local storage
3. Session storage

Module 10 - HTML Media

- introduction
- Audio
- Vedio
- Accessing audio/video from external source file in webpage

Module 11 - HTML Graphics

- Introduction to SVG and Canvas
- Creating line , circle , rectangle and polygon etc using svg /

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CSS 3 :

Module 12- Introduction

- Version history
- Css syntax
- Styling using external style sheets

Module 13 - Selectors

- Specificity
- Tag selector • ID Selectors
- Class Selectors
- Grouping Selectors
- Universal Selector
- Descendant / Child Selectors
- Attribute Selectors
- CSS – Pseudo Classes **Module 14 - Text Fonts**
- color
- background-color
- text-decoration
- text-align
- vertical-align
- text-indent
- text-transform
- white space
- letter-spacing
- word-spacing
- line-height
- font-family
- font-size
- font-style
- font-variant
- font-weight

Module 15 - Background

- Background Image
- Background color
- Background Position
- background repeat

Module 16 - Border

- order width
- Border Type
- Border Color
- Border radius

Module 17 - Box Model

- Margin
- Padding
- Width
- Height
- Borders
- Outline

Module 18 - Positioning

- Static
- Fixed
- relative • Absolute
- Z-index

Module 19 - Display

- nline
- Inline Block
- block
- Table
- Flex
- Grid
- NONE

Module 20 - Visibility

- Visible
- Hidden

Module 21 - Responsive Web Design:

- Media Queries
- Breaking Points
- Targeting Web Page

- Targeting PDF

Module 22 - Transforms

- 2D Transforms
- 3D Transforms
- Transitions
- Animations
- CSS Short Hand
- Box Shadow
- Gradient
- Flex Box
- HTML Graphics

Module 52: React JS Installation

- Introduction to React JS
- Setting up the development environment
- Installing React using npm or yarn
- Creating a new React app with Create React App (CRA)
- Running the development server

Module 53: Basics

- Understanding React and its core concepts
- Virtual DOM vs Real DOM
- Components in React
- Rendering elements
- Importance of single-page applications (SPAs)

Module 54: JSX

- Introduction to JSX (JavaScript XML)
- Writing HTML-like syntax in JavaScript
- Embedding expressions in JSX
- JSX attributes and styling
- JSX vs plain JavaScript

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Module 55:

Class Components & Functional Components

- Understanding components in React
- Class components: structure and usage
- Functional components: modern approach
- Differences between class and functional components
- When to use each type

Module 56: Component Props

- What are props (properties)?
- Passing data between components
- Accessing props in class and functional components
- Default props
- Props vs state

Module 57: Component State

- Introduction to component state
- Setting and updating state
- Using state in class components (this.state)
- Using React hooks (useState) in functional components
- State vs props

Module 58- Hooks

- For
- For in
- While
- Do While
- For of

Module 59- Communication between components.

- For
- For in
- While
- Do While
- For of

Module 60- Reusable Components

- For
- For in

- While
- Do While
- For of

Module 61- Working with API data using fetch / axios library

- For
- For in
- While
- Do While
- For of

Module 62- Integrating Prime React UI Component library.

- For
- For in
- While
- Do While
- For of

Module 63- React Router

- Creating Routes and Router Layouts
- Navigation from JSX and React components
- Child Routes
- Reading Router Params
- Routing Context
- Authentication for routes. **Module 64- React Forms**
- Creating Forms using vanilla react● Creating forms using formic library.

Module 65- Redux

- Store
- Actions
- Dispatchers
- Reducers
- Immutability
- State management using redux library.

Module 66- Redux vs Flux

- Store
- Actions
- Dispatchers
- Reducers
- Immutability
- State management using the redux library.

Module 67- Unit Testing

- Understanding testing
- Introduction to Jest
- Test Running using Jest
- Mocking functions and modules
- Testing Components● Snapshot Testing
- Async / Await.

PYTHON

Module 1: Introduction to Python

- What is Python? Why Python? Features & Applications Installing
- Python & Setting Up Environment (VS Code, Jupyter Notebook)
- Running Python Scripts (CMD & IDEs)

Module 2: Basic Python Concepts

- Variables and Data Types
- Operators in Python (Arithmetic, Logical, Bitwise, etc.)
- Type Conversion and Casting

- Input and Output Functions

Module 3: Control Flow Statements

- Conditional Statements (if, elif, else)
- Looping Statements (for, while)
- Loop Control Statements (break, continue, pass)

Module 4: Data Structures in Python

- Lists
- Tuples
- Dictionaries
- Sets

Module 5: Functions & Modules

- Defining and Calling Functions
- Function Arguments (Default, Keyword, Arbitrary Arguments)
- Lambda Functions
- Built-in Functions
- Importing and Creating Modules

Module 6: Working with Strings

- String Operations
- String Slicing
- String Methods
- Regular Expressions (re module)

Module 7: File Handling

- Reading and Writing Files
- Working with CSV Files
- Exception Handling in File Operations

Module 8: Exception Handling

- Understanding Exceptions
- Try, Except, Finally Blocks
- Raising Exceptions

Module 9: Object-Oriented Programming (OOPs)

- Classes and Objects Constructors (init method) Inheritance
- (Single, Multiple, Multilevel, Hybrid) Polymorphism & Method
- Overriding Encapsulation & Data Hiding
- List Comprehensions
- Generators & Iterators

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MYSQL

Module 1: Introduction to MySQL

- What is MySQL? Features of MySQL
- Installing MySQL (Windows, Linux) MySQL
- Workbench & Command Line Interface
- Understanding Databases and Tables

Module 2: MySQL Basics

- SQL vs MySQL Data
- Types in MySQL
- Creating a Database (CREATE DATABASE)
- Selecting a Database(USE DATABASE)
- Deleting a Database (DROP DATABASE)

Module 3: Working with Tables

- Creating Tables (CREATE TABLE)
- Column Constraints (NOT NULL, UNIQUE, DEFAULT, CHECK)
- Primary Key & Foreign Key Altering Tables (ALTER TABLE)
- Deleting Tables (DROP TABLE)

Module 4: CRUD Operations

- INSERT Data (INSERT INTO)
- READ/SELECT Data (SELECT FROM)
- UPDATE Data (UPDATE SET WHERE)
- DELETE Data (DELETE FROM WHERE)

Module 5: Filtering Data

- WHERE Clause Using AND, OR, NOT
- Operators BETWEEN, IN, LIKE, IS
- NULL Operators Sorting Data
- (ORDER BY ASC | DESC) Limiting
- Results (LIMIT)

Module 6: Aggregate Functions & Grouping Data

- COUNT, SUM, AVG, MIN, MAX
- Functions Grouping with GROUP BY Filtering
- Groups with HAVING

Module 7: MySQL Joins

- INNER JOIN LEFT JOIN (LEFT
- OUTER JOIN) RIGHT JOIN
- (RIGHT OUTER JOIN) FULL
- JOIN (FULL OUTER JOIN)
- Self Join
- Cross Join

Module 8: Subqueries & Nested Queries

- What is a Subquery?
- Single Row Subqueries
- Multiple Row Subqueries
- Correlated Subqueries Using EXISTS
- and NOT EXISTS

Module 9: Views and Indexing

- Creating Views (CREATE VIEW) Updating &
- Deleting Views (UPDATE, DROP VIEW) What is
- Indexing? Creating Indexes (CREATE INDEX)
- Types of Indexes (Primary, Unique, Full-text)

Module 10: Stored Procedures & Functions

- Creating a Stored Procedure (CREATE PROCEDURE)
- Calling a Stored Procedure (CALL)
- MySQL Functions (CREATE FUNCTION)
- Differences Between Procedures & Functions
- Triggers & Events What are Triggers?
- Creating Triggers (BEFORE INSERT, AFTER INSERT, BEFORE UPDATE, **AFTER UPDATE)**
- Deleting Triggers MySQL Events & Scheduling Tasks
- Understanding Transactions Using COMMIT & ROLLBACK
  Savepoints in Transactions (SAVEPOINT, ROLLBACK TO SAVEPOINT)

Module 13: Window functions

- Window functions Types of window
- functions Numeric functions Rank
- functions
- Installing Django
- Starting a Django project
- Running the Django server
- Exploring the project files

Django

Setting your development environment:

Creating your first Django project:

- Creating a Django app
- Adding a page
- When it doesn't work
- Django and the flow of control
- Adding an about page

Django's MVT architecture:

- Introducing templates and the MVT pattern
- A template for the welcome page
- Template variables and dynamic content
- Completing the MVT pattern
- Taking a parameter from the URL
- Returning a 404 error

Data models:

- Create read updates and delete records
- Define relationships between data
- Validate data before saving
- Query database using ORM
- Introducing models and migrations
- Running initial migrations
- Creating model class
- Creating and running migrations

- The admin interface

Views:

- Http requests (get, post, put delete, options.)
- Processing data from models
- Handle user inputs
- Form validation
- Returning appropriate https responses(html, Json, Redirects)

Templates:

- Introduction to DTL(django template language)
- Inserting data dynamically into HTML
- Template inheritance
- Filters
- Tags
- Adding styling with CSS
- Adding an image

Django forms:

- A template with a modelform
- Submitting the form
- Processing user input
- Understanding the program flow for processing a form
- Validation and security

URL Routing:

- Url configuration
- Url pattern matching
- Path converters
- Naming URL patterns
- Url name spaces
- Class based views
- List and detailed views
- Debugging url configurations **Admin interface:**
- Basic setup
- Custom model admin class
- Field sets for forms

- In line editing for models
- Custom display methods with formatting
- Custom filtering ad search

Authentication:

- Basic setup
- Configuring authentication middleware
- Registration template
- Login and logout
- Password reset
- Protecting views
- User context

Vibe coding

MODULE 1 — Vibe Coding Fundamentals

- What is Vibe Coding?
- Tools Overview
- How to Prompt for Code
- Vibe Debugging
- Building with Cursor (Composer Mode)
- Git & Version Control (AI-Assisted)
- Deploying Your First Webpage

MODULE 2 — Frontend Vibe Coding

- HTML Structure via Prompts
- CSS Styling through AI
- Responsive Design
- AI-Generated Components
- Animations & Effects
- Full Frontend Page Build
- Frontend Review + Deployment

MODULE 3 — Backend Foundations

- What is Backend?
- Node.js Express via Vibe Coding
- CRUD Operations
- Database Introduction (Supabase)
- Connect Backend to DB
- Testing APIs via AI
- Backend Clean-Up

MODULE 4 — Full-Stack Basics

- Connecting Frontend & Backend
- Authentication (Simple Login)
- Login UI + JWT Flow
- State Management Basics
- Error Handling via AI
- Full-Stack To-Do App
- Deployment (Full Stack)

MODULE 5 — Intermediate Features

- Pagination, Filters, Sorting
- File Uploads
- Email API Integration
- Notifications & Automation
- Third-Party API Integration
- Analytics + Logs
- Security Basics

MODULE 6 — Advanced Vibe Coding

- AI-Generated Database Schema
- Role-Based Access
- Payments Integration (Demo Mode)
- Building Admin Dashboard
- Automation Workflows
- Refactoring Big Projects
- Mid-Bootcamp Review

MODULE 7 — Real-World Projects

- E-commerce Product Page
- Shopping Cart System
- Orders + Checkout Flow
- Blog System
- Portfolio Generator
- Event Booking App
- Chat-Based Task Manager

MODULE 8 — Career & Final Project

- Pick Your Final Project
- Plan the Architecture
- Build Frontend
- Build Backend
- Integrations
- UI Polish + Testing
- Deploy + Portfolio

Tools

- Cursor

- Lovable
- ChatGPT

Projects

- Implementing conference application
- Creating a meeting
- Editing a meeting
- Deleting a meeting
- Display all meeting using grids and charts
- Enabling authentication for users
- Implementing top level common templates
- Configuring login and logout views
- Reacting to user login status
- Restricting access to views
- Customizing the admin interface